

Vaani Bhatnagar

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EDUCATION

Olin College of Engineering

Needham, MA | May 2025 | 3.8/4.0

Bachelor's of Science in Mechanical Engineering

Relevant Coursework: Collaborative Design, Mechanical Design, Principles of Integrated Engineering, Sustainable Design, Software Design, Design for Manufacturing, Accessibility in Design

SKILLS

Design: Figma, InDesign, Photoshop, Illustrator, Product Design, Mechanical Design, Human-Centered Design, System Integration, Miro, User Research

Technical Tools: ReactJS, Python, Java, C/C++, Azure, SQL, Jira, Confluence, SolidWorks, Arduino, MATLAB,

Product Management: Agile Methodologies, Stakeholder Collaboration, Supply Chain Optimization

EXPERIENCE

Product Designer - Optum (Basking Ridge, NJ)

July 2025 - Present

- Collaborate with healthcare stakeholders to translate complex medical requirements into clear, actionable design solutions through user interviews and iterative design testing.
- Develop prototypes and design systems for healthcare applications, ensuring consistency across enterprise products, creating intuitive workflows for Electronic Medical Records and clinical tools to reduce cognitive load for clinicians and enhancing patient care quality.

UI/UX Designer - Microsoft NERD | Olin Senior Capstone Project (Needham, MA)

September 2024 - May 2025

- Co-managed a project to design and prototype a consent framework for health data that empowers marginalized communities, while collaborating with multiple liaisons as the communications lead, integrating user feedback into iterative design cycles, and creating scalable user research frameworks and practices through focus groups and codesigns to make evidence-based design decisions and build trust with the community.

Product Design Intern - Optum (Basking Ridge, NJ)

June 2022 - August 2022

- Led product research and competitor analysis to identify integration opportunities for 5 Optum products, presenting a multi-product model to the CTO that improved governance and customer satisfaction.
- Utilized Jira to manage projects in Agile workflows.

Operations Intern - Accelerate Wind (Remote)

September 2024 - December 2024

- Conducted market and user research and crafted grant proposals to secure funding for renewable energy solutions, integrating rooftop solar panels and wind turbines, while supporting product management, business planning, supply chain strategy, and sourcing to scale the adoption of innovative energy systems.

Mechanical Design Engineer - NASA Nationwide Eclipse Ballooning Project (Needham, MA)

December 2023 - May 2024

- Designed and tested resilient enclosures for high-altitude balloon missions, ensuring durability in extreme conditions including 70,000+ ft altitudes, 120+ mph winds, and unpredictable landing orientations, safeguarding critical circuitry and cameras.

Software Engineering Intern - Optum (Basking Ridge, NJ)

June 2024 - August 2024

- Designed and implemented a CI/CD data pipeline with Apache Kafka, GitHub Actions, AWS S3, and Java in an Agile/Scrum environment, automating ETL tool testing to enhance system security and performance, achieving a \$0.5 million boost.

Data Science Intern - Optum (Basking Ridge, NJ)

June 2023 - August 2023

- Built a Snowflake Database that improved patient query speed by 100x and reduced costs by 79%, leveraging SQL, Python, and Azure Data Factory to automate data flow using Agile practices.

PROJECTS

FAIR Consent Framework (User Interviews, User Journey Mapping, Interactive Prototyping, Iterative Design Testing)

- Designed a user-centered consent framework for health data sharing, translating complex privacy and compliance requirements into clear, step-by-step user flows; improved user comprehension of consent choices and reduced cognitive load during data-sharing decisions as measured through iterative usability testing and qualitative feedback through focus groups.

The Pawfect Match (User Persona and Empathy Mapping, Design System Development, Wireframing, Usability Testing)

- Led end-to-end design of a matching platform for animal adoption, conducting user research with shelter staff and adopters to design empathetic matching flows that improved perceived fit quality, reduced mismatches, and supported shelter operational efficiency based on user testing and stakeholder feedback.